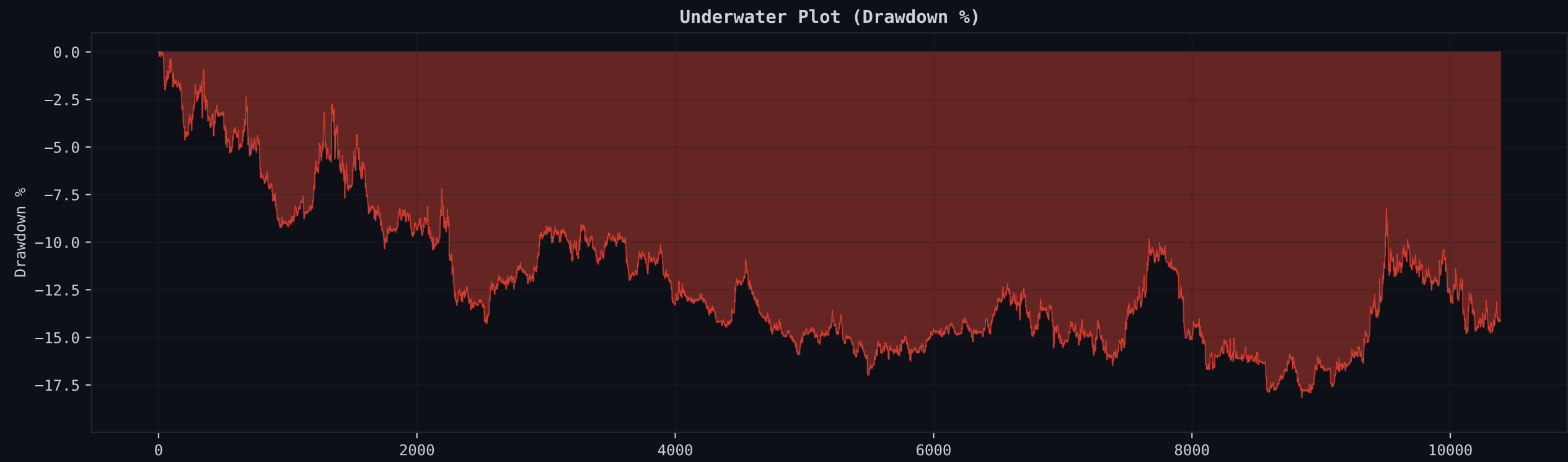
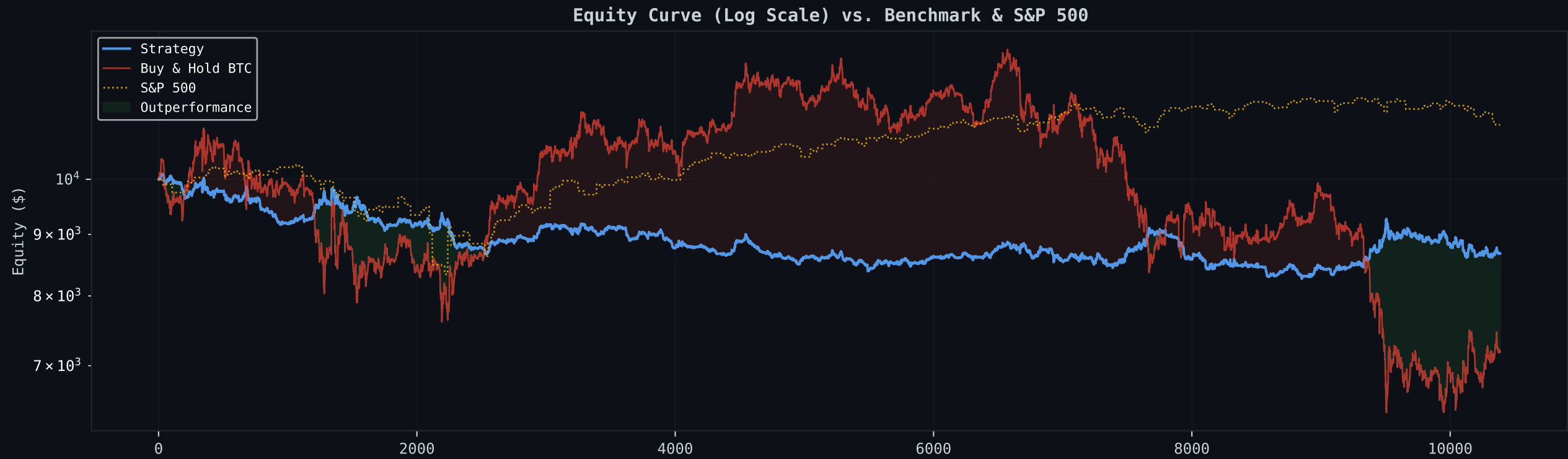
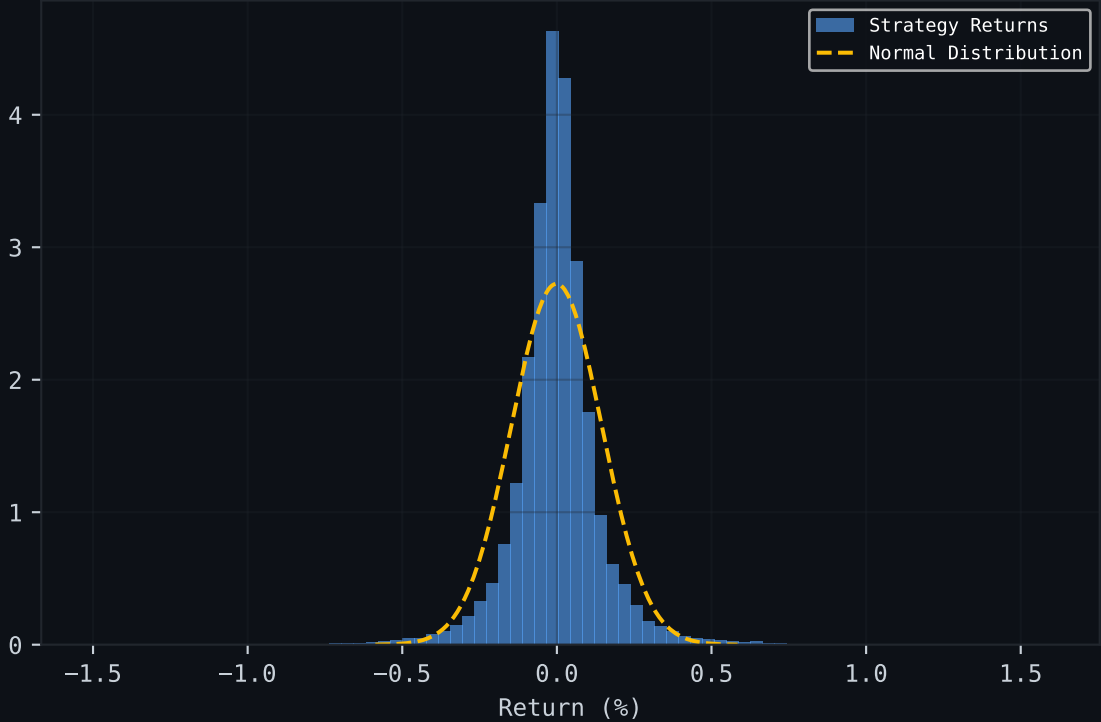


INSTITUTIONAL QUANTITATIVE TEARSHEET
CNN+LSTM PPO × SPA (Pure OOS Walk-Forward)

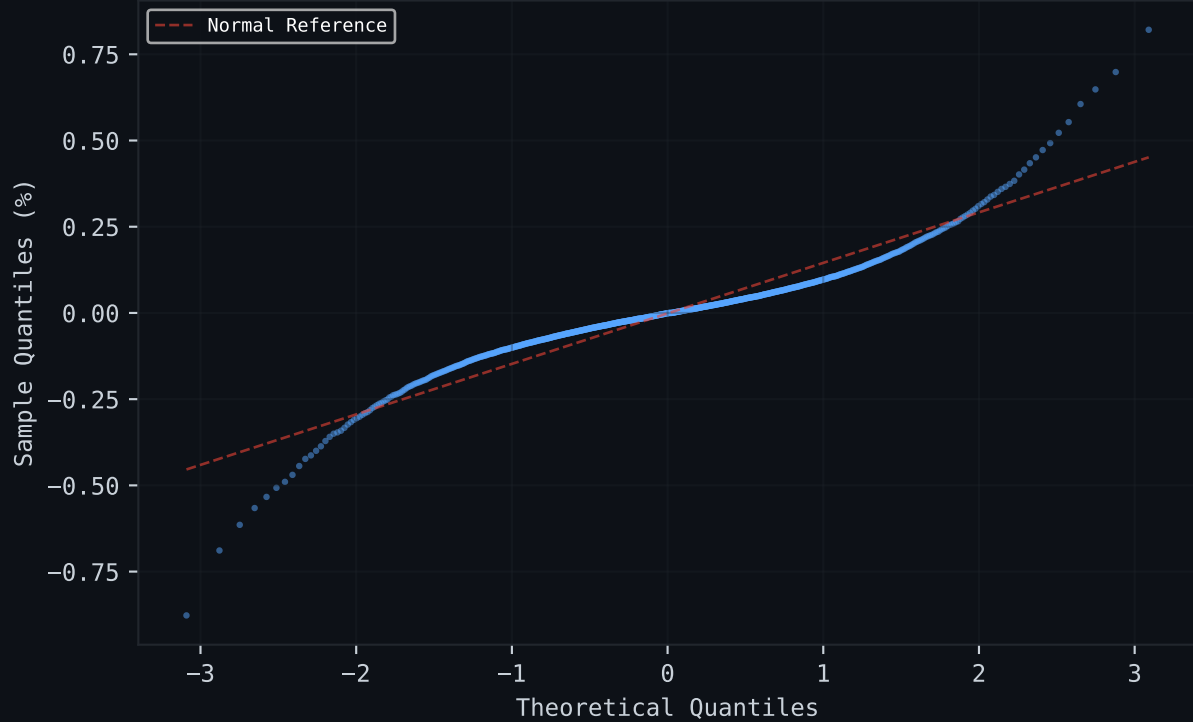
Sharpe: -0.82 | Sortino: -1.08 | Calmar: -0.62 | Max DD: -18.2% | Alpha p-value: 0.8197 | Beta: -0.035



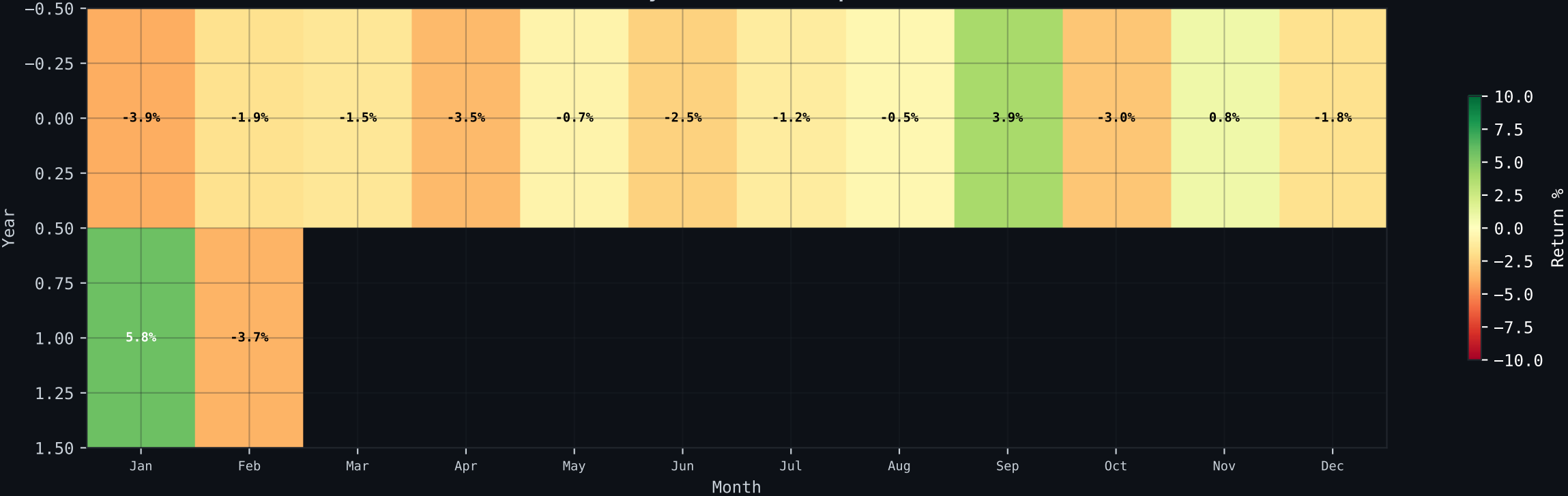
Return Distribution vs. Normal



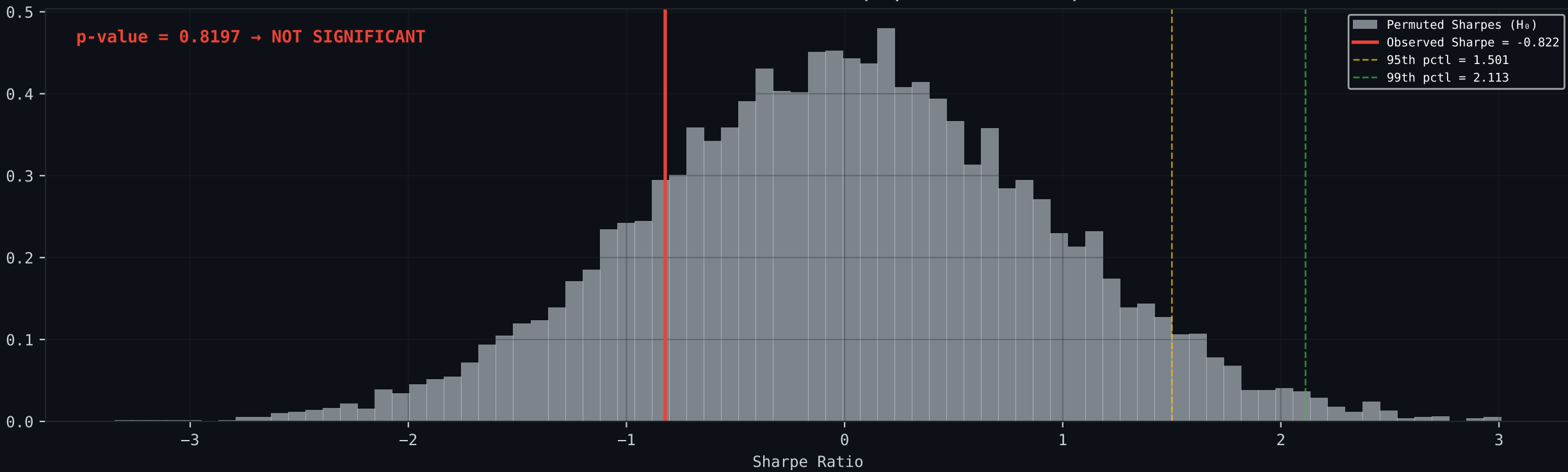
Q-Q Plot (Fat Tail Detection)



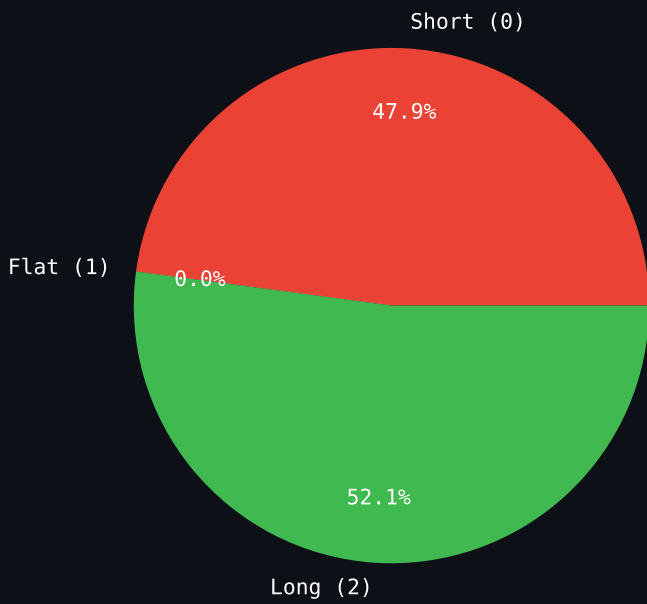
Monthly Returns Heatmap (%)



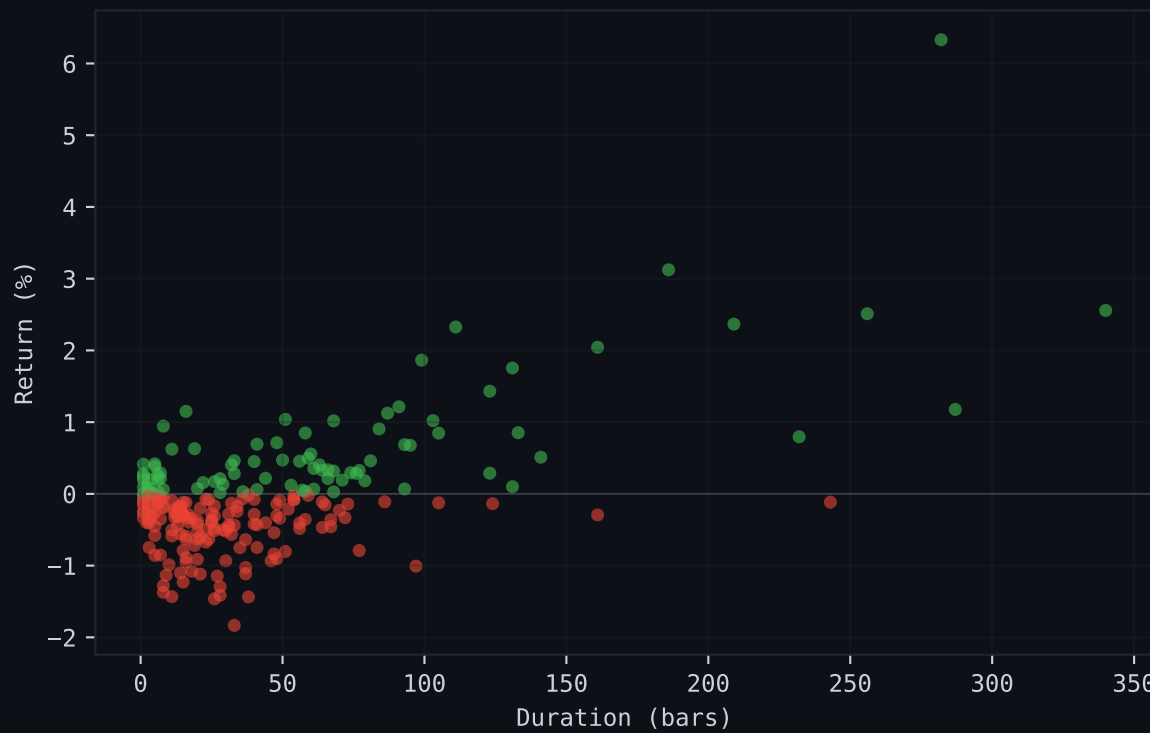
Monte Carlo Permutation Test (10,000 simulations)



Action Distribution



Trade Returns vs. Duration



QUANTITATIVE RISK METRICS – FULL DETAIL

Metric	Value
PERFORMANCE METRICS	
Total Return	-13.22%
Annualized Return	-11.27%
Annualized Volatility	13.71%
Sharpe Ratio	-0.822
Sortino Ratio	-1.078
Calmar Ratio	-0.620
Omega Ratio (θ=0)	0.974
Profit Factor	0.974
Win Rate	49.3%
BENCHMARK COMPARISON	
BTC Buy & Hold Sharpe	-0.527
BTC Buy & Hold Sortino	-0.671
BTC Buy & Hold Max DD	-50.10%
S&P 500 Sharpe	0.517
S&P 500 Sortino	0.112
S&P 500 Max DD	-18.90%
TAIL RISK METRICS	
Max Drawdown	-18.17%
VaR (95%)	-0.2112%
VaR (99%)	-0.4231%
CVaR / Expected Shortfall (95%)	-0.3504%
CVaR / Expected Shortfall (99%)	-0.6112%
Skewness	-0.070
Excess Kurtosis	11.630
ALPHA DECOMPOSITION	
Alpha (per period)	-0.000013
Alpha t-statistic	-0.929
Alpha p-value	0.3529
Beta (benchmark exposure)	-0.0346
R² (benchmark correlation)	0.0136
STATISTICAL VALIDATION	
Permutation Test p-value	0.8197
Significance (p < 0.05)	NO ☐
CAPACITY ESTIMATION	
Est. Capacity (USD)	\$0
Current Market Impact	0.29 bps
Participation Rate	0.000009%